PHB 227
Statistical Learning and Multivariate Analysis

MODEL SELECTION

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Motivation

Given several predictors, how do we choose the "best" subset?
Why would we want to do this?

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Why would we want to do this?

- Inference/Mechanisms
- Prediction
- Cost
- More predictors than subjects ($p > n$).

Model Selection Methods

- Subset selection: best subset, stepwise (forward selection, backward elimination, hybrid)
- Shrinkage (aka Regularization)
- Dimension Reduction: project the p predictors onto a low dimensional space; use the projections as predictors.

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We will cover **Shrinkage** in depth.

Shrinkage Methods

Assume we have a generalized linear model with p predictors: $g(Ey_i) = \sum_{j=1}^{p} x_{ij}\beta_j$ with link function g, and associated log-likelihood function, l.

- Ridge Regression:
  Find β that minimizes

$$-\sum l(y_i, \beta; x_i) + \lambda \sum \beta_j^2$$

  or equivalently
  Find β that minimizes

$$-\sum l(y_i, \beta; x_i) \text{ subject to } \sum \beta_j^2 \leq s$$

  What are the indices for the two sums?
- Lasso:
  Find β that minimizes

$$-\sum l(y_i, \beta; x_i) + \lambda \sum |\beta_j|$$

  or equivalently
  Find β that minimizes

$$-\sum l(y_i, \beta; x_i) \text{ subject to } \sum |\beta_j| \leq s$$

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When $\lambda = 0$, there is no shrinkage; as $\lambda \to \infty$, the coefficients shrink to 0.

For each λ we get a set of coefficients; we need to choose "optimal" λ.

Ridge and lasso are not invariant to scaling so the predictors are usually scaled before applying these procedures, i.e., replace x_{ij} by $z_{ij} = \frac{x_{ij}}{s_j}$, where $s_j^2 = \frac{\sum_i (x_{ij} - \bar{x}_{\cdot j})^2}{n-1}$. We usually also center the predictors, so $z_{ij} = \frac{x_{ij} - \bar{x}_{\cdot j}}{s_j}$, where $\bar{x}_{\cdot j} = \frac{\sum_i x_{ij}}{n}$.

Henceforth in the theoretical framework, we will generally assume the predictors and outcome are centered and scaled.

Example: Heart data

```
xnams2

## [1] "Age"    "RestBP" "Chol"   "MaxHR"

xxx <- as.matrix(Heart[,xnams2])
yyy <- Heart$AHD

gridd <- exp(seq(2,-6, -0.5))  ##lambda values

##ridge fit
rdg.fit <- glmnet(xxx,yyy,family="binomial",alpha=0,lambda=gridd)

##lasso fit
lso.fit <- glmnet(xxx,yyy,family="binomial",alpha=1,lambda=gridd)

##lambda
#gridd

rdg.fit$lambda

##  [1] 7.389056099 4.481689070 2.718281828 1.648721271
##  [5] 1.000000000 0.606530660 0.367879441 0.223130160
##  [9] 0.135335283 0.082084999 0.049787068 0.030197383
## [13] 0.018315639 0.011108997 0.006737947 0.004086771
## [17] 0.002478752
```

```
##very shrunk
rdg.fit$lambda[1]

## [1] 7.389056

coef(rdg.fit)[,1][xnams2]

##           Age        RestBP          Chol         MaxHR
##  0.0015609801  0.0005490558  0.0001047999 -0.0011832721

coef(lso.fit)[,1][xnams2]

##    Age RestBP   Chol  MaxHR
##      0      0      0      0

##logistic coeff: no shrinkage!
glm(fmla2, data=Heart,family="binomial")$coef[xnams2]

##           Age        RestBP          Chol         MaxHR
##  0.004931645  0.016496078  0.003126728 -0.043286860
```

```r
##almost no shrinkage
rdg.fit$lambda[length(gridd)]

## [1] 0.002478752

coef(rdg.fit)[,length(gridd)][xnams2]

##          Age       RestBP         Chol        MaxHR
##  0.005664347  0.016152007  0.003056061 -0.042439863

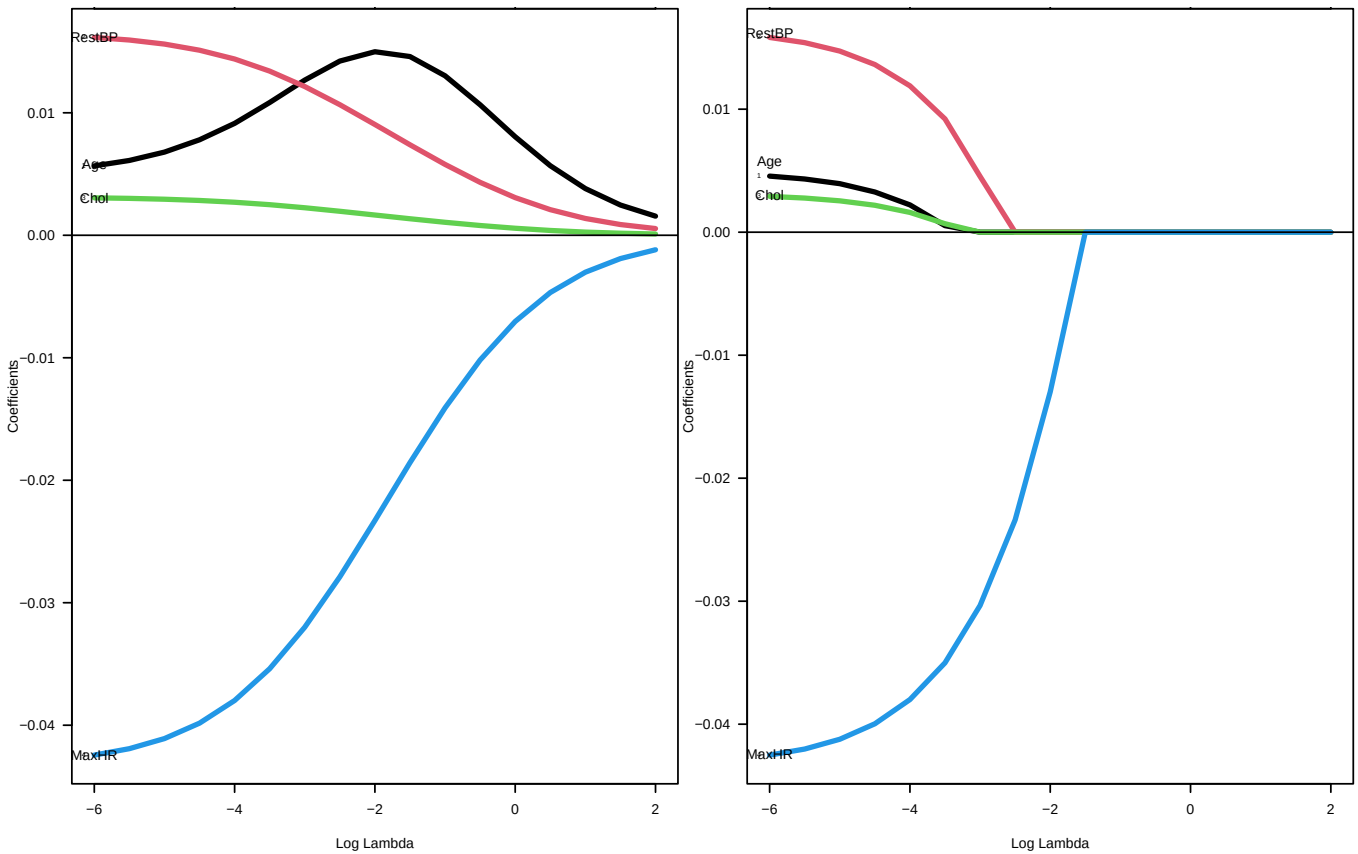
coef(lso.fit)[,length(gridd)][xnams2]

##          Age       RestBP         Chol        MaxHR
##  0.004559838  0.015833590  0.002911756 -0.042512522

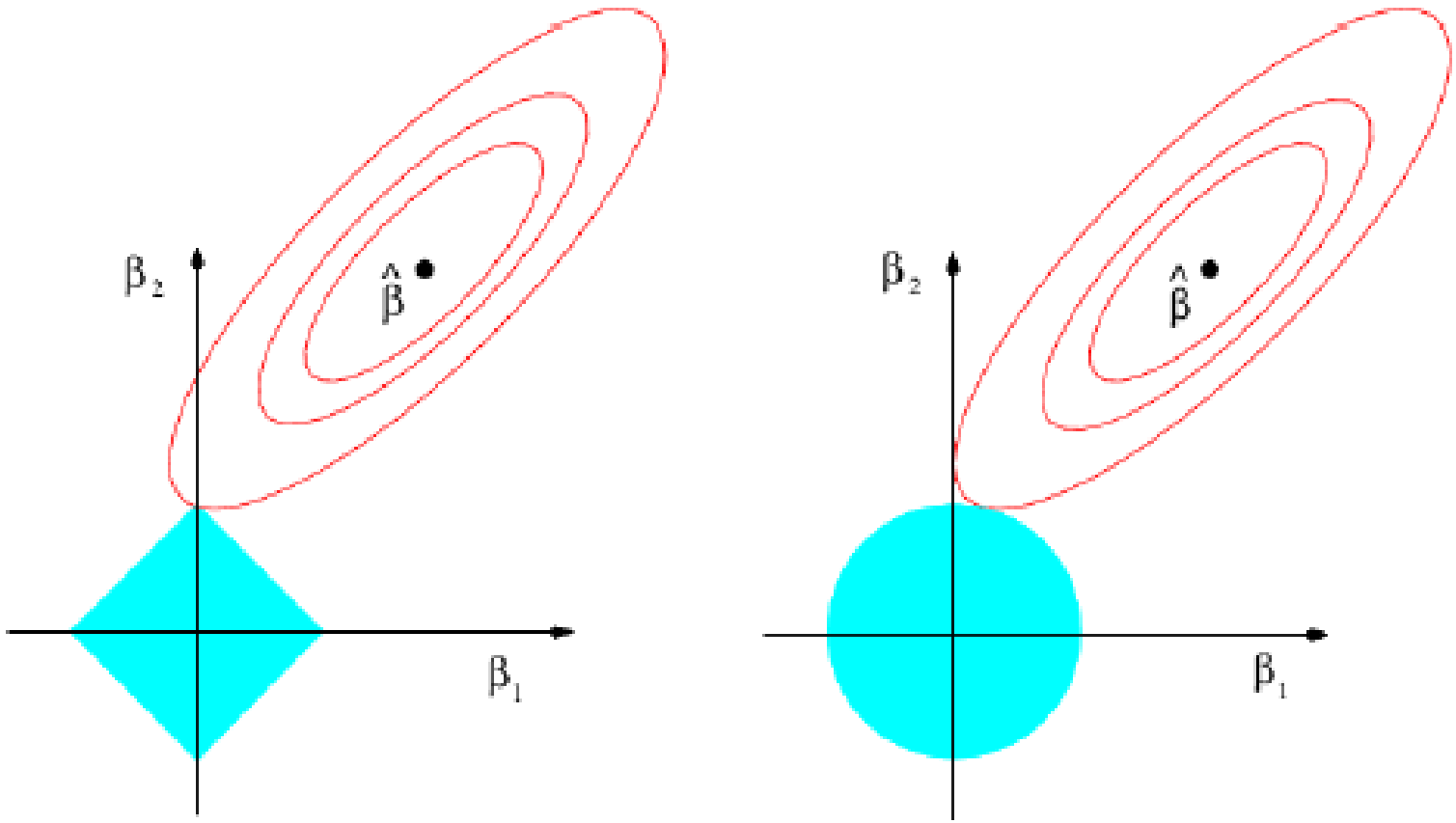
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##          Age       RestBP         Chol        MaxHR
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```

```
## [1] "Age"    "RestBP" "Chol"   "MaxHR"
```



Variable selection property of LASSO



Lasso (left) and ridge regression (right): The blue areas are the constraint regions, $|\beta_1| + |\beta_2| \leq s$ and $\beta_1^2 + \beta_2^2 \leq s$; red ellipses are the contours of the RSS (i.e., RSS = constant). Due to the shape of the constraints on the coefficients, β, LASSO sets some coefficients to 0.

Prediction with Ridge and Lasso

See Rscript

How to choose λ

Cross-validation to the rescue!

1. Choose a grid of values for λ
2. Split sample into training and held-out sets (e.g., 10-fold)
3. For each λ, fit shrinkage model on training set
4. Test fitted model on held-out set, and compute error (squared-loss, deviance)
5. Repeat 2-4 and get cross-validated errors for each λ
6. Select λ for which error is lowest

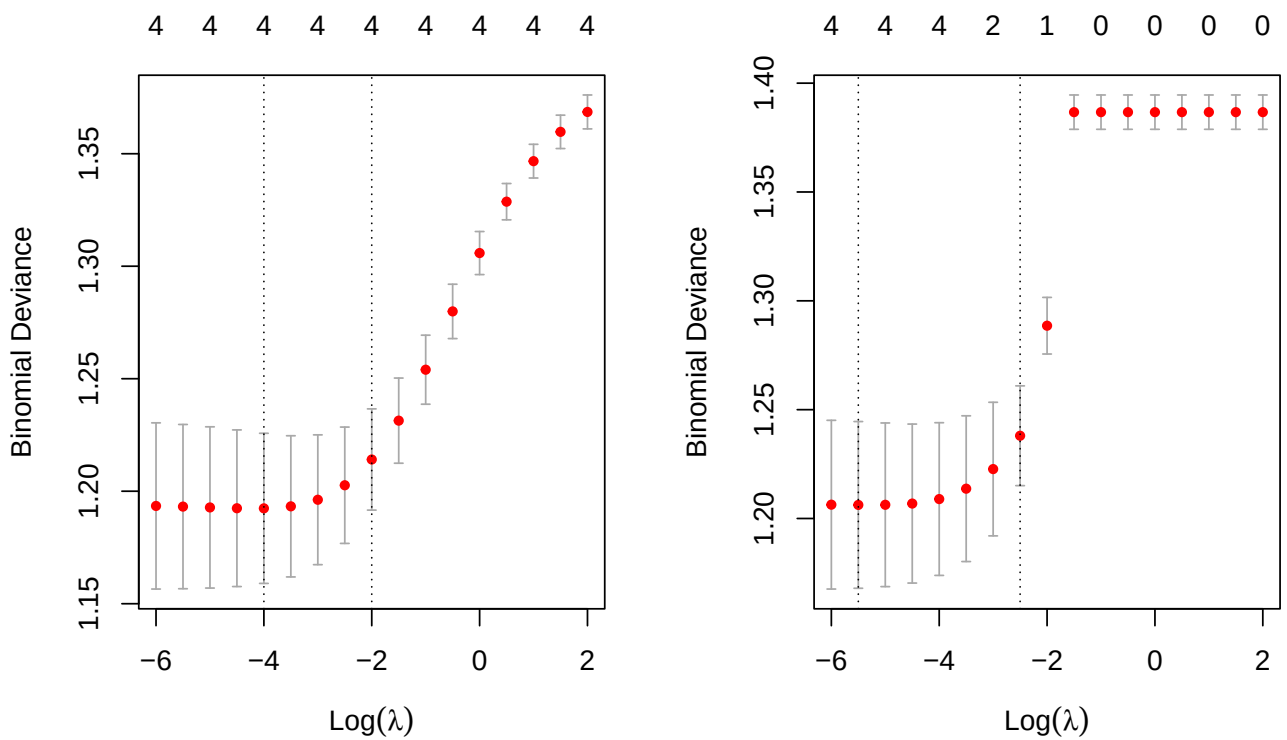
Once λ is selected, we refit the model on the full dataset with this choice of λ.

How to choose λ: Heart dataset

```
##cross-validation to select lambda
set.seed(2446)
cv.rdgeg <- cv.glmnet(xxx,yyy,family="binomial",alpha=0,lambda=gridd,nfolds=10)

cv.lsoeg <- cv.glmnet(xxx,yyy,family="binomial",alpha=1,lambda=gridd,nfolds=10)
log(c(cv.rdgeg$lambda.min,cv.lsoeg$lambda.min))

## [1] -4.0 -5.5
```



Variables (and coefficients) at optimal λs for RIDGE

```
##Coeff at "best" lambda
coef(cv.rdgeg,s="lambda.min")

## 5 x 1 sparse Matrix of class "dgCMatrix"
##                       s1
## (Intercept)  2.444764151
## Age          0.009135779
## RestBP       0.014390278
## Chol         0.002696933
## MaxHR       -0.037981406

##Coeff at  lambda + 1SE
coef(cv.rdgeg,s="lambda.1se")

## 5 x 1 sparse Matrix of class "dgCMatrix"
##                       s1
## (Intercept)  0.892534644
## Age          0.014986190
## RestBP       0.009045126
## Chol         0.001652794
## MaxHR       -0.023289228
```

Variables (and coefficients) at optimal λs for LASSO:

```
##Coeff at "best" lambda
coef(cv.lsoeg,s="lambda.min")

## 5 x 1 sparse Matrix of class "dgCMatrix"
##                       s1
## (Intercept)  3.159828452
## Age          0.004325199
## RestBP       0.015410434
## Chol         0.002773839
## MaxHR       -0.042020103

##Coeff at  lambda + 1SE
coef(cv.lsoeg,s="lambda.1se")

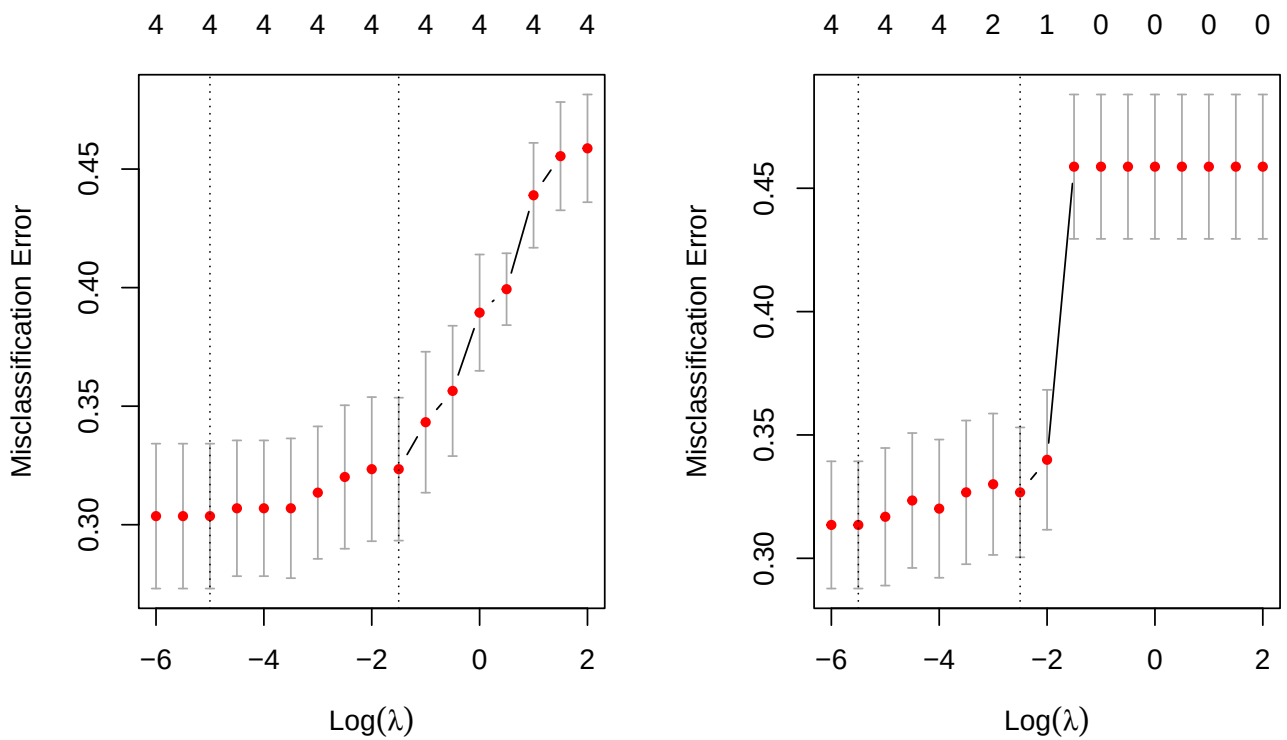
## 5 x 1 sparse Matrix of class "dgCMatrix"
##                      s1
## (Intercept)  3.32845908
## Age          .
## RestBP       .
## Chol         .
## MaxHR       -0.02338832
```

Using misclassification as the loss function

```
##cross-validation to select lambda
set.seed(2446)
cv.rdgeg2 <- cv.glmnet(xxx,yyy,family="binomial",alpha=0,lambda=gridd,nfolds=10

cv.lsoeg2 <- cv.glmnet(xxx,yyy,family="binomial",alpha=1,lambda=gridd,nfolds=10
log(c(cv.rdgeg2$lambda.min,cv.lsoeg2$lambda.min))

## [1] -5.0 -5.5
```



Linear Case: OLS, Ridge, Lasso

Consider again the linear model

$$Y = X\beta + \epsilon$$

and assume Y and X are centered.

$$\text{Suppose } \text{rank}(X) < p.$$

Let us consider the implications of this for OLS, Ridge and Lasso....

Why OLS does not work

If $\text{rank}(X) < p$, then the nullspace of X is non-empty. This implies that the OLS solution is not unique. As a result:

- Interpretation of coefficients is ambiguous.
  Why?

- Prediction is not unique.
  Why?

Note: Fitted values for OLS *are* unique.

Importantly, the above problems of OLS are resolved by the Ridge and Lasso methods.

Ridge Solution: Linear Case

In linear regression, the objective for ridge regression is to find $\beta(\lambda)$ that minimizes

$$\frac{||Y - X\beta||^2}{2} + \lambda||\beta||^2.$$

The solution to the ridge optimization problem is given by

$$\hat{\beta}^{ridge}(\lambda) = (X'X + 2\lambda I_p)^{-1}X'Y.$$

Let us derive this.....
Comment: Using the singular value decomposition of X, one can show that ridge shrinks most those X- directions (in the column space of X) which have the least variance.

Lasso Solution: Linear Case

Lemma: Let $G(\beta)$ be the gradient of $\frac{||Y-X\beta||^2}{2}$. A necessary and sufficient condition for $\hat{\beta}$ to be a solution to the lasso optimization problem for linear regression is

$$\begin{aligned} G_j(\hat{\beta}) &= -sign(\hat{\beta}_j)\lambda \text{ if } \hat{\beta}_j \neq 0 \\ |G_j(\hat{\beta})| &\leq \lambda \text{ if } \hat{\beta}_j = 0. \end{aligned}$$

Also, if the lasso solution is not unique (e.g., if $p > n$) and $G_j(\hat{\beta}) < \lambda$ for some solution $\hat{\beta}$, then $\hat{\beta}_j = 0$ for all lasso solutions.

See Buhlmann, van de Geer for proof and other details.

Note: It is also true that any two lasso solutions will have the same sign on the overlap of their support (support is the set of indices of non-zero coefficients).

In general there is no closed form for the lasso solution. Let us consider the orthogonal X case....

Ridge vs Lasso: Linear regression with Orthogonal X

Assume we are in the linear regression setting. Also, suppose the predictor matrix X is orthogonal, i.e., $X^T X = I$, the identity matrix. Now consider the least-squares solution given by $\hat{\beta}_j = X^T Y$. Under these assumptions

- RIDGE: $\hat{\beta}_j^{ridge} = \frac{\hat{\beta}_j}{1+2\lambda}$
- LASSO: $\hat{\beta}_j^{lasso} = \text{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \lambda)_+$.

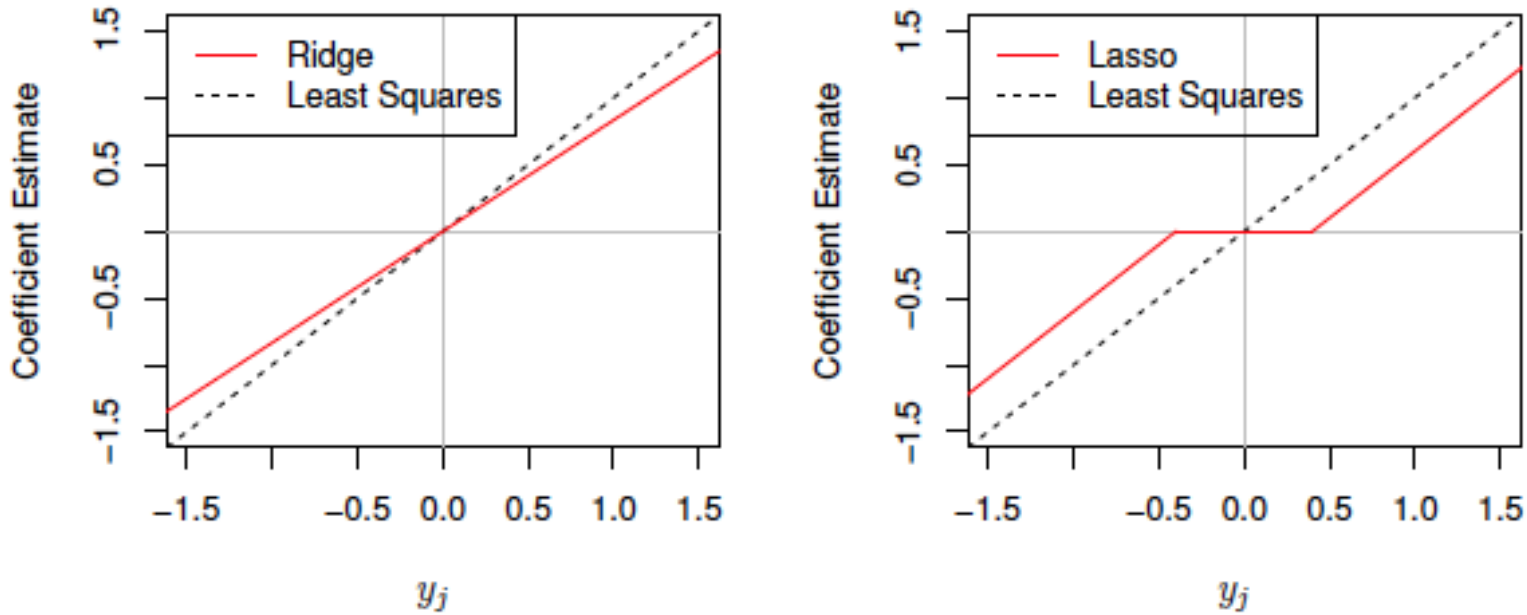


FIGURE 6.10. Simple setting with n = p and X identity matrix. Left: The ridge regression coefficient estimates are shrunken proportionally towards zero, relative to the least squares estimates. Right: The lasso coefficient estimates are soft-thresholded towards zero.

Comparing Methods: Simulation

Assume a linear model

$$Y = \beta_0 + \beta' X + \text{ error },$$

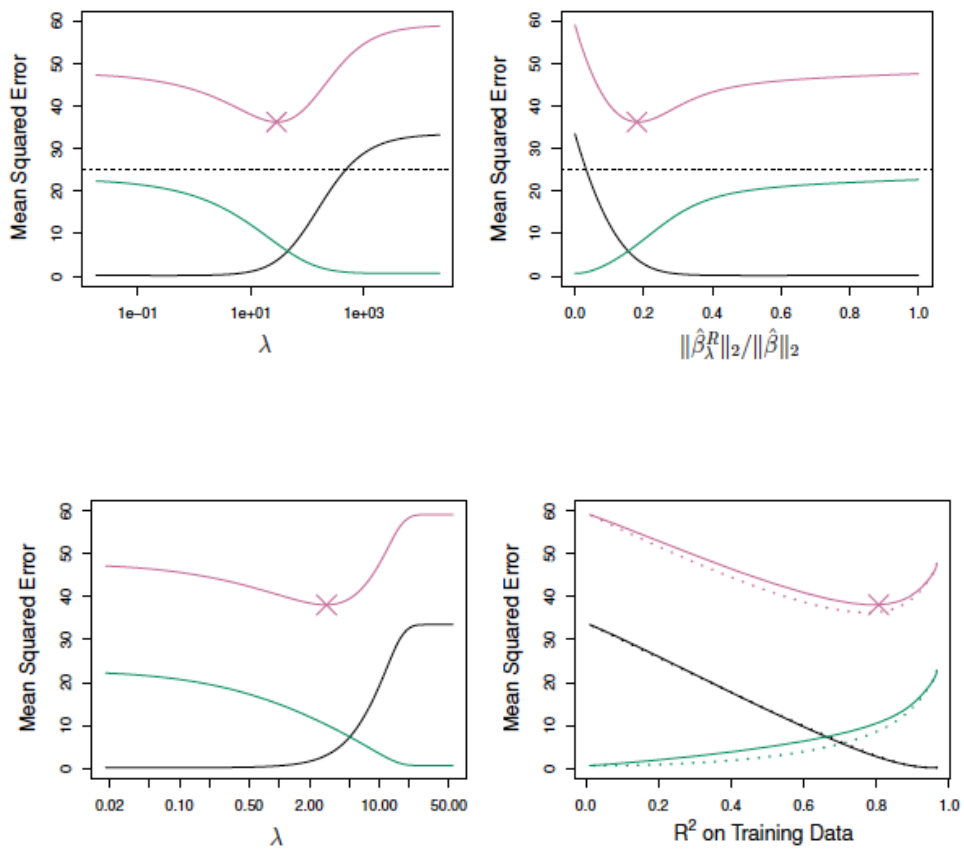
so loss function is squared-error loss $(y - \beta_0 - \beta' x)^2$
We will generate Gaussian X (multivariate) and error (scalar), and generate Y as above. In the simulations, $n = 100, p = 90$

- Example 1: $X_1, X_2, \cdots, X_p$ are independent Gaussian with mean 0, variance 1; $\beta_1 = \beta_2 = 10$; $\beta_3 = \beta_4 = 5$; $\beta_5 = \beta_6 = 1$; $\beta_j = 0$ for $j > 6$.
- Example 2:$X_1, X_2, \cdots, X_p$ are independent Gaussian with mean 0, variance 1; $\beta_j = 1$ for $j = 1, 2, \cdots, 90$.

Compare the performance of ridge and lasso methods in the above simulations.

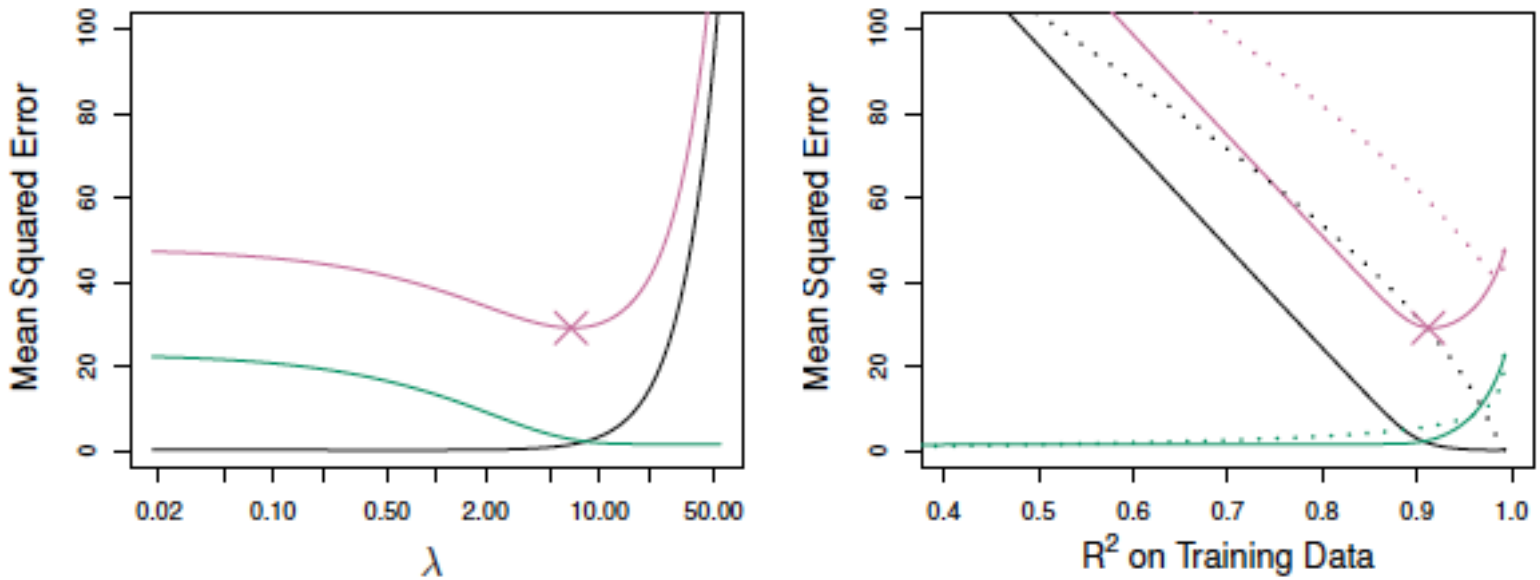
Comparing Methods: ISLR example Figures 6.5, 6.8

Ridge (top, bottom right-dotted) and lasso (bottom) squared bias (black), variance (green), and test MSE (purple) on a simulated data set containing $p = 45$ predictors and $n = 50$ observations. All 45 predictors have non-zero coefficients in the simulated dataset. The crosses indicate the model with smallest MSE.



Comparing Methods: ISLR Figure 6.9

Ridge and lasso performance on a simulated data set containing $p = 45$ predictors and $n = 50$ observations. Only 2 of the 45 predictors have non-zero coefficients in the simulated dataset.



Left: Squared bias (black), variance (green), and test MSE (purple) for the lasso. Right: Comparison between lasso (solid) and ridge (dotted). Both are plotted against their R2 on the training data. The crosses indicate the lasso model with smallest MSE.

Comparing Methods: Broad conclusions

- Compared to least-squares, lasso and ridge introduce some bias, but reduce variance
- Neither ridge nor lasso is universally better than the other
  - Lasso will generally do better in sparse situations where there are a few predictors with large effects and the rest have negligible effects
  - Ridge will generally do better when there are many predictors with similar effects
- Lasso usually performs variable selection, i.e., sets some coefficients to 0
- In any given application, use cross-validation to choose a method

Elastic Net

Elastic Net combines the Ridge (L_2) and Lasso (L_1) penalties:

Find β that minimizes

$$-\sum l(y_i, \beta; x_i) + \lambda_1 \sum \beta_j^2 + \lambda_2 \sum |\beta_j|$$

The elastic net method includes the Lasso and ridge regression as special cases. How?

The quadratic penalty term makes the loss function strongly convex, and it therefore has a unique minimum.

If there is a group of highly correlated variables, then the Lasso tends to select one variable from a group and ignore the others. Elastic Net addresses this drawback of Lasso.

Zou, H.,Hastie,T (2005) JRSSB

Bayesian interpretation

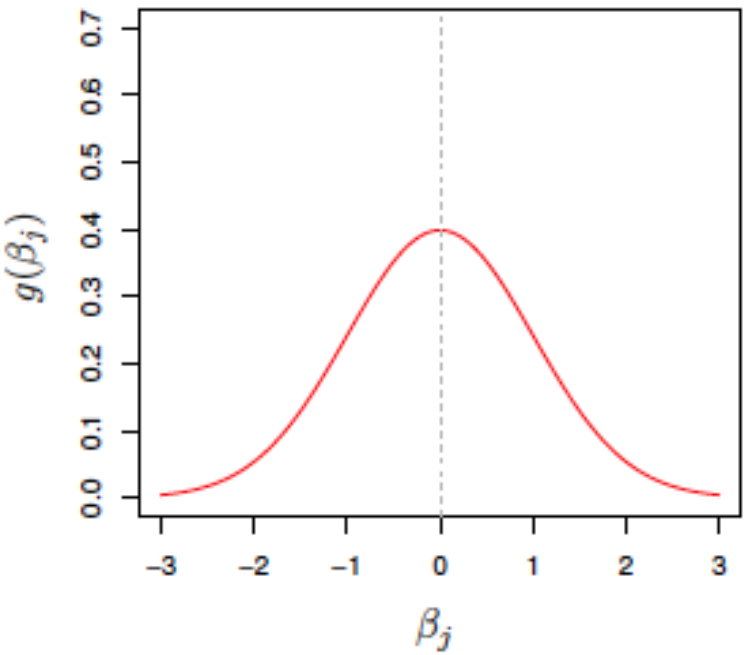
Suppose $\beta = (\beta_1, \beta_2, \cdots, \beta_p)$, and let $f(Y|X, \beta)$ be the likelihood of Y given $X = (X_1, X_2, \cdots, X_p)$. By Bayes theorem, the posterior density of β given X, Y is

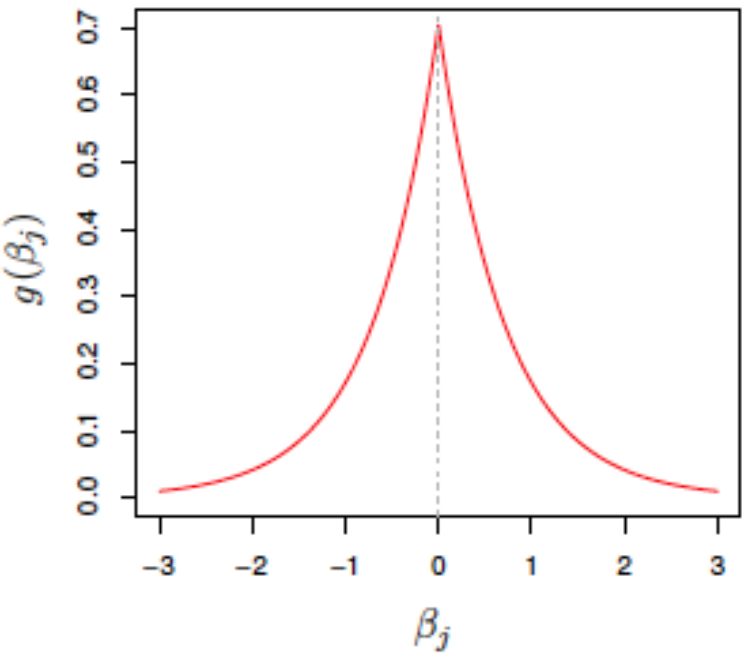
$$p(\beta|X, Y) \propto f(Y|X, \beta)p(\beta|X) = f(Y|X, \beta)\pi(\beta),$$

where π is the prior density of β.
Now assume $Y = X\beta + \epsilon$, i.e., a linear model with independent, mean-zero, Gaussian errors and suppose the prior $\pi(\beta) = \Pi_{j=1}^{p} g(\beta_j)$ for some density g,

- Ridge: If g is Gaussian with mean zero and standard deviation a function of λ, then the posterior mode for β given the data, is given by the ridge regression solution.
- Lasso: If g is double exponential with mean zero and scale parameter a function of λ, then the posterior mode for β given the data, is the lasso regression solution.

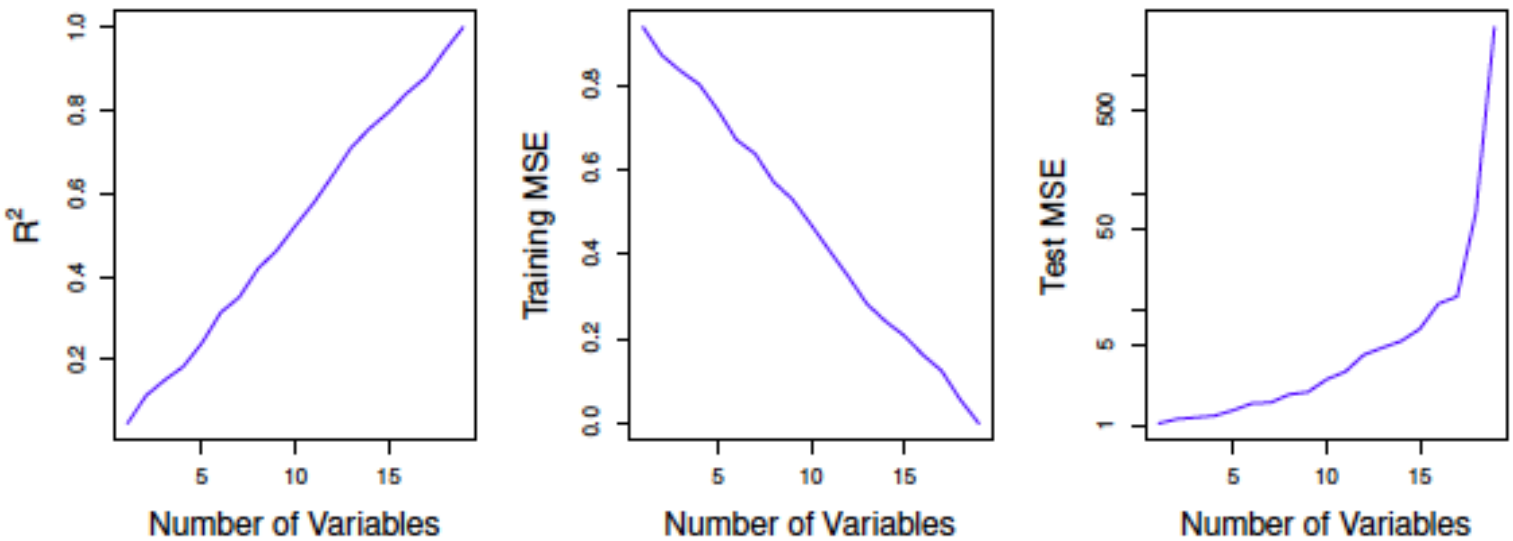
Bayesian interpretation





High-dimensional setting: Cautionary example

When the # of predictors p is close to or larger than n, then least-squares (or other standard methods) will overfit.
ISLR Fig. 6.23 (below) shows simulated data (n = 20) with upto 20 predictors where predictors X and response Y are unrelated.



High-dimensional setting: Cautionary example

- What do you notice in the left and middle panels? What might you conclude?
- What do you notice in the right hand panel?
- Hpw do you explain the discrepancy between the left/middle and right panels?

High-dimensional setting

When the # of predictors p is close to or larger than n, then least-squares (or other standard methods) will overfit. In fact, if $p > n$ there is no unique solution, and we may get a perfect fit to the training set, but have very high variance on test sets.

Shrinkage methods can be applied, but may still need to be interpreted cautiously.

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See Leng, Lin, Wahba Statistica Sinica 2006.
Adaptive Lasso and SCAD attempt to address this "inconsistency", i.e., attempt to find the "true" variables.

ISLR Example Fig 6-24

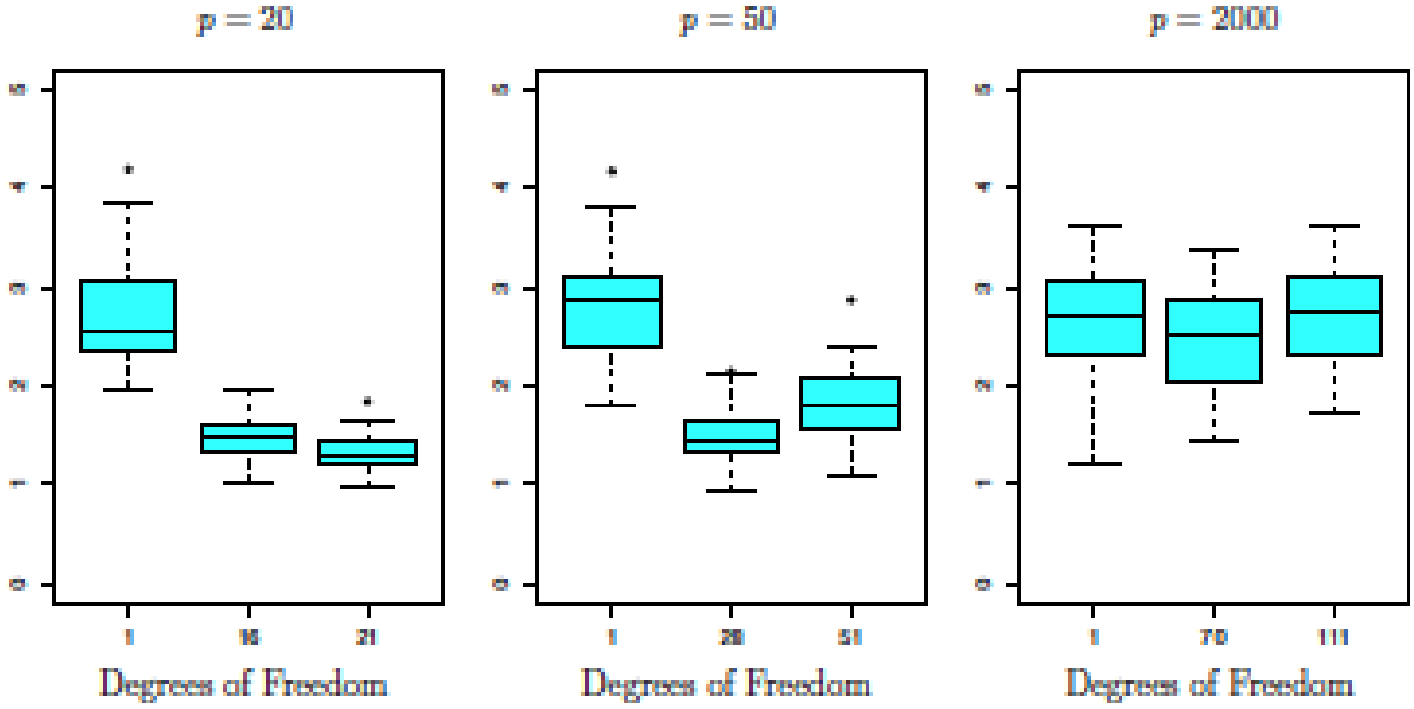


FIGURE 6.24. *The lasso was performed with $n = 100$ observations and three values of p, the number of features. Of the p features, 20 were associated with the response. The boxplots show the test MSEs that result using three different values of the tuning parameter λ in (6.7). For ease of interpretation, rather than reporting λ, the* degrees of freedom *are reported; for the lasso this turns out to be simply the number of estimated non-zero coefficients. When $p = 20$, the lowest test MSE was obtained with the smallest amount of regularization. When $p = 50$, the lowest test MSE was achieved when there is a substantial amount of regularization. When $p = 2{,}000$ the lasso performed poorly regardless of the amount of regularization, due to the fact that only 20 of the 2,000 features truly are associated with the outcome.*

Other Penalties

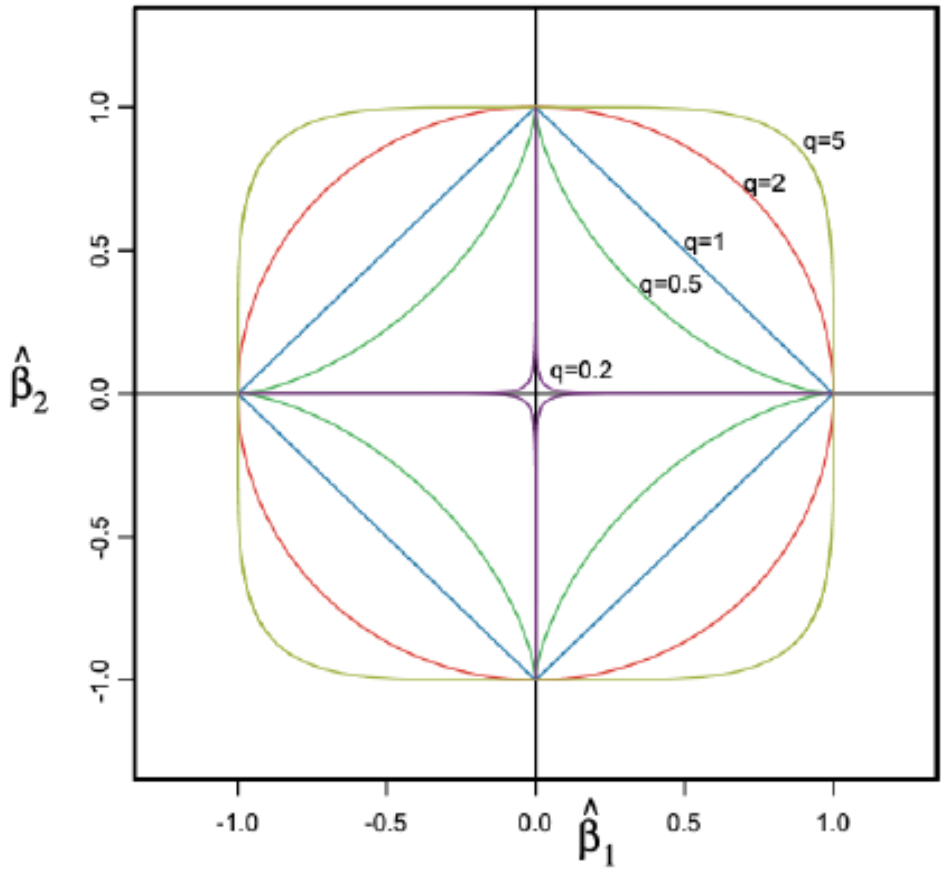


FIGURE 5.10. *Two-dimensional contours of the symmetric penalty function* $p_q(\boldsymbol{\beta}) = |\beta_1|^q + |\beta_2|^q = 1$ *for* $q = 0.2, 0.5, 1, 2, 5$. *The case* $q = 1$ *(blue diamond) yields the lasso and* $q = 2$ *(red circle) yields ridge regression.*

Izenman book (2008)

Review Questions

- How is the Ridge regression problem specified?
- How is the Lasso regression problem specified?
- What is the relationship between the size of the penalty and the size of the coefficients?
- What is the coefficient path for ridge or lasso regressions?
- How is the "optimal" penalty determined?
- What are the problems with applying OLS when the design matrix is ill-coditioned?
- In the orthogonal case, describe the ridge solution.
- In the orthogonal case, describe the lasso solution.
- What is a Bayesian interpretation of the ridge and lasso problems.
- In general, in which situations would you expect lasso to outperform ridge regression? Conversely, when would you expect ridge to outperform lasso?